

**Definition 4.** *The process defined on the interval  $[0, 1]$  as*

$$X(t) := \sum_{j=0}^{+\infty} \sum_{k=0}^{2^j-1} \left( \int_0^1 (t-s)_+^{H_j(k/2^j)-1/2} h_{j,k}(s) ds \right) \varepsilon_{j,k}, \quad (6)$$

*will be called the Gaussian Haar-based multifractional process. The abbreviation GHBMP will be used to denote processes from this class.*

Note that the GHBMP processes defined by (6) are centered Gaussian, but do not have stationary increments.

### 3. On GHBMP and its Hölder continuity

This section studies GHBMP and its basic properties. It also provides several key characteristics of Haar-based transforms that will be used in subsequent analyses.

The first result shows that the series in Definition 4 is  $L_2$ -convergent and GHBMP is correctly defined.

**Theorem 3.1.** *For all fixed  $t \in [0, 1]$  it holds*

$$\mathbb{E}X^2(t) = \sum_{j=0}^{+\infty} \sum_{k=0}^{2^j-1} \left| \int_0^1 (t-s)_+^{H_j(k/2^j)-1/2} h_{j,k}(s) ds \right|^2 < +\infty.$$

*Therefore, the GHBMP processes are properly defined in  $L_2$ -sense.*

The proof of Theorem 3.1 relies on the following result.

**Lemma 1.** *For any fixed value  $\lambda \in [\underline{H}, \overline{H}]$ , let  $h^{[\lambda]}(\cdot)$  be the function defined on  $\mathbb{R}$  as*

$$\begin{aligned} h^{[\lambda]}(x) &:= \int_{\mathbb{R}} (x-s)_+^{\lambda-\frac{1}{2}} h(s) ds \\ &= \left( \lambda + \frac{1}{2} \right)^{-1} \left( (x)_+^{\lambda+\frac{1}{2}} - 2 \left( x - \frac{1}{2} \right)_+^{\lambda+\frac{1}{2}} + (x-1)_+^{\lambda+\frac{1}{2}} \right). \end{aligned} \quad (7)$$

*Then, there is a constant  $c > 0$  such that for all  $(\lambda, x) \in [\underline{H}, \overline{H}] \times \mathbb{R}$ ,*

$$|h^{[\lambda]}(x)| \leq c(3 + |x|)^{\lambda-\frac{3}{2}}. \quad (8)$$